

Question number	Answer	Additional guidance	Mark
3(a)(i)	2772	IGNORE any other units given	(1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	An explanation that includes four of the following points: • light is absorbed by {photosystems / chlorophyll} (1) • which {excites electrons / releases high-energy electrons / releases electrons to higher energy levels} (1) • these electrons are passed along a series of (electron) carriers (1) • therefore releasing <u>energy</u> to phosphorylate ADP into ATP (cyclic)(1) • phosphorylation of ADP via the proton gradient to form ATP (non-cyclic) (1)	ACCEPT description e.g. hydrogen ions pass through ATP synthase releasing energy for phosphorylation of ADP NB reference to ATP being synthesised from ADP only needed once to award both 4th and 5th marking point	(4)

Question number	Answer	Additional guidance	Mark
3(b)(i)	two from: C - H, C - O and C - C	IGNORE O - H ACCEPT bond between carbon and hydrogen bond between carbon and oxygen bond between carbon and carbon	(1)

Question number	Answer	Mark
3(b)(ii)	The only correct answer is C. <i>A is incorrect because there is no cytoplasm inside chloroplasts</i> <i>B is incorrect because the matrix is not found in chloroplasts</i> <i>D is incorrect because glucose is synthesized in the stroma of chloroplasts</i>	(1)

Question number	Answer	Mark
3(c)(i)	carbon and nitrogen condensation The only correct answer is A. <i>B is incorrect because bonds form by condensation reactions not hydrolysis</i> <i>C is incorrect because the peptide bond joins the C of one amino acid to the N of the other</i> <i>D is incorrect because the peptide bond joins the C of one amino acid to the N of the other and bonds form by condensation reactions not hydrolysis</i>	(1)

Question number	Answer	Additional guidance	Mark
3(c)(ii)	An explanation that includes two of the following points: - because amino acids contain nitrogen - because some {amino acids / R groups} contain sulfur - nitrogen obtained from nitrates / sulfur obtained from sulfates	(1) ACCEPT glucose does not contain nitrogen IGNORE nitrates ACCEPT glucose does not contain sulfur IGNORE sulfates ACCEPT nitrates / sulfates needed	(2)